

Inner Product Spaces

- Inner product:

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in a vector space V , and let c be any scalar. An inner product on V is a function that associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ with each pair of vectors

$$\mathbf{u} = (u_1, u_2, \dots, u_n) \quad , \quad \mathbf{v} = (v_1, v_2, \dots, v_n)$$

and satisfies the following axioms.

- (1) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
- (2) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
- (3) $c \langle \mathbf{u}, \mathbf{v} \rangle = \langle c\mathbf{u}, \mathbf{v} \rangle$
- (4) $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$

- Note:

$\mathbf{u} \cdot \mathbf{v}$ = dot product (Euclidean inner product for R^n)

$\langle \mathbf{u}, \mathbf{v} \rangle$ = general inner product for vector space V

Vector space: $(V, +, \bullet)$

Inner product space: $(V, +, \bullet, \langle, \rangle)$

- Ex 6.1: (The Euclidean inner product for R^n)

Show that the dot product in R^n satisfies the four axioms of an inner product.

Sol:

$$\mathbf{u} = (u_1, u_2, \dots, u_n) \quad , \quad \mathbf{v} = (v_1, v_2, \dots, v_n)$$

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

By Def., this dot product satisfies the required four axioms. Thus it is an inner product on R^n .

Weighted Euclidean Inner Product

The Euclidean inner product is the most important inner product on $\mathbb{R}^n$. However, there are various applications in which it is desirable to modify the Euclidean inner product by *weighting its terms differently*. More precisely, if

$$w_1, w_2, \dots, w_n$$

are positive real numbers, which we shall call **weights**, and if $\mathbf{u} = (u_1, u_2, \dots, u_n)$

and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ are vectors in $\mathbb{R}^n$, then it can be shown that the formula

$$\langle \mathbf{u}, \mathbf{v} \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n$$

defines an inner product on $\mathbb{R}^n$; it is called the **weighted Euclidean inner product with weights**

$$w_1, w_2, \dots, w_n$$

■ Ex 6.2: (A different inner product for R^n)

Show that the function defines an inner product on R^2 , where

$$\mathbf{u} = (u_1, u_2) \quad \mathbf{v} = (v_1, v_2)$$

and $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2$

Sol:

$$(a) \quad \langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2 = v_1 u_1 + 2v_2 u_2 = \langle \mathbf{v}, \mathbf{u} \rangle$$

$$(b) \quad \mathbf{w} = (w_1, w_2)$$

$$\begin{aligned} \Rightarrow \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= u_1(v_1 + w_1) + 2u_2(v_2 + w_2) \\ &= u_1 v_1 + u_1 w_1 + 2u_2 v_2 + 2u_2 w_2 \\ &= (u_1 v_1 + 2u_2 v_2) + (u_1 w_1 + 2u_2 w_2) \\ &= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle \end{aligned}$$

$$(c) \quad c \langle \mathbf{u}, \mathbf{v} \rangle = c(u_1 v_1 + 2u_2 v_2) = (cu_1)v_1 + 2(cu_2)v_2 = \langle c\mathbf{u}, \mathbf{v} \rangle$$

$$(d) \quad \langle \mathbf{v}, \mathbf{v} \rangle = v_1^2 + 2v_2^2 \geq 0$$

$$\langle \mathbf{v}, \mathbf{v} \rangle = 0 \Rightarrow v_1^2 + 2v_2^2 = 0 \quad \Rightarrow \quad v_1 = v_2 = 0 \quad (\mathbf{v} = \mathbf{0})$$

Since all four axioms of inner product are satisfied, therefore, the given function is an inner product space for $\mathbb{R}^2$.

■ Ex 6.3: (A function that is not an inner product)

Show that the following function is not an inner product on R^3 .

$$\langle \mathbf{u} \cdot \mathbf{v} \rangle = u_1 v_1 - 2u_2 v_2 + u_3 v_3$$

Sol:

Let

$$\mathbf{v} = (1, 2, 1)$$

$$\text{Then } \langle \mathbf{v}, \mathbf{v} \rangle = (1)(1) - 2(2)(2) + (1)(1) = -6 < 0$$

Axiom 4 is not satisfied.

Thus this function is not an inner product on R^3 .

- Theorem 6.1: (Properties of inner products)

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in an inner product space V , and let c be any real number.

$$(1) \quad \langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$$

$$(2) \quad \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(3) \quad \langle \mathbf{u}, c\mathbf{v} \rangle = c \langle \mathbf{u}, \mathbf{v} \rangle$$

- Norm (length) of $\mathbf{u}$:

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$$

- Note:

$$\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle$$

- Distance between $\mathbf{u}$ and $\mathbf{v}$:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}$$

- Angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi$$

- Orthogonal: $(\mathbf{u} \perp \mathbf{v})$

$$\mathbf{u} \text{ and } \mathbf{v} \text{ are orthogonal if } \langle \mathbf{u}, \mathbf{v} \rangle = 0$$

■ Notes:

(1) If $\|\mathbf{v}\| = 1$, then $\mathbf{v}$ is called a unit vector.

(2) $\|\mathbf{v}\| \neq 1$ $\xrightarrow{\text{Normalizing}}$ $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ (the unit vector in the direction of $\mathbf{v}$)
 $\mathbf{v} \neq \mathbf{0}$
not a unit vector

■ Ex 6.4: (Finding inner product)

$\langle p, q \rangle = a_0b_0 + a_1b_1 + \cdots + a_nb_n$ is an inner product

Let $p(x) = 1 - 2x^2$, $q(x) = 4 - 2x + x^2$ be polynomials in $P_2(x)$

(a) $\langle p, q \rangle = ?$ (b) $\|q\| = ?$ (c) $d(p, q) = ?$

Sol:

$$(a) \quad \langle p, q \rangle = (1)(4) + (0)(-2) + (-2)(1) = 2$$

$$(b) \quad \|q\| = \sqrt{\langle q, q \rangle} = \sqrt{4^2 + (-2)^2 + 1^2} = \sqrt{21}$$

$$(c) \quad \because p - q = -3 + 2x - 3x^2$$

$$\begin{aligned} \therefore d(p, q) &= \|p - q\| = \sqrt{\langle p - q, p - q \rangle} \\ &= \sqrt{(-3)^2 + 2^2 + (-3)^2} = \sqrt{22} \end{aligned}$$

■ Properties of norm:

(1) $\|\mathbf{u}\| \geq 0$

(2) $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$

(3) $\|c\mathbf{u}\| = |c| \|\mathbf{u}\|$

■ Properties of distance:

(1) $d(\mathbf{u}, \mathbf{v}) \geq 0$

(2) $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$

(3) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$

■ Theorem 6.2 :

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V .

(1) Cauchy-Schwarz inequality:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

(2) Triangle inequality:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

(3) Pythagorean theorem :

$\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Orthonormal Bases

- Orthogonal:

A set S of vectors in an inner product space V is called an **orthogonal set** if every pair of vectors in the set is orthogonal.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subseteq V$$

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \quad i \neq j$$

- Orthonormal:

An orthogonal set in which each vector is a unit vector is called **orthonormal**.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subseteq V$$

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- Note:

If S is a basis, then it is called an orthogonal basis or an orthonormal basis.

- Ex 6.5: (A nonstandard orthonormal basis for R^3)

Show that the following set is an orthonormal basis.

$$S = \left\{ \overset{\mathbf{v}_1}{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)}, \quad \overset{\mathbf{v}_2}{\left(-\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3} \right)}, \quad \overset{\mathbf{v}_3}{\left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right)} \right\}$$

Sol:

Show that the three vectors are mutually orthogonal.

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -\frac{1}{6} + \frac{1}{6} + 0 = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = \frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} + 0 = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = -\frac{\sqrt{2}}{9} - \frac{\sqrt{2}}{9} + \frac{2\sqrt{2}}{9} = 0$$

Show that each vector is of length 1.

$$\|\mathbf{v}_1\| = \sqrt{\mathbf{v}_1 \cdot \mathbf{v}_1} = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1$$

$$\|\mathbf{v}_2\| = \sqrt{\mathbf{v}_2 \cdot \mathbf{v}_2} = \sqrt{\frac{2}{36} + \frac{2}{36} + \frac{8}{9}} = 1$$

$$\|\mathbf{v}_3\| = \sqrt{\mathbf{v}_3 \cdot \mathbf{v}_3} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1$$

Thus S is an orthonormal set.